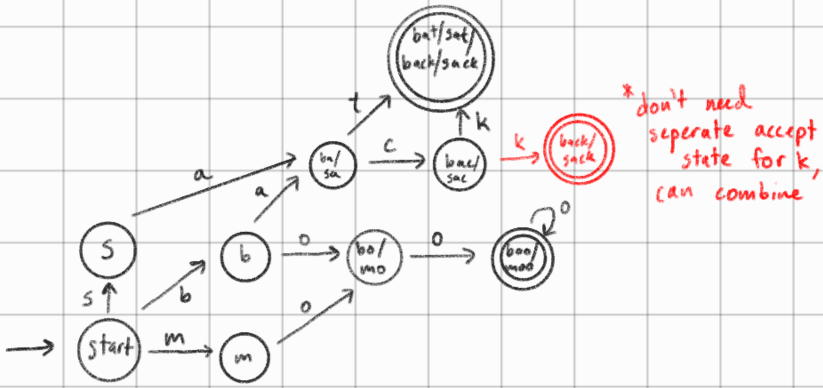
CS 173 - Lecture 25

More State Diagrams

ex. Phone lattice (deterministic) that accepts bat, back, sat, sack, boo, booo, boooo... (boo w/ two or more o's), moo, mooo, moooo... (moo w/ two or more o's)

strat: look for shared suffixes/prefixes

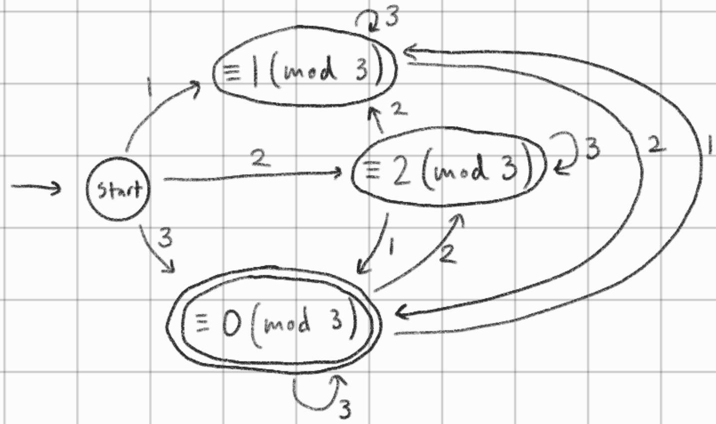
Assume missing transitions go to dump state



ex. build deterministic state diagram that takes sequence of 1's, 2's, and 3's and enters an accept state whenever number read so far is multiple of 3 (base 10).

sum of digits is multiple of 3

need $\equiv 0 \pmod 3$ or $\equiv 3 \pmod 3$, only possible sums are $\equiv 0 \pmod 3$, $\equiv 1 \pmod 3$, $\equiv 2 \pmod 3$



Discussion Problems – Magic Word

State Diagrams Additional Tutorial Problem

Magic Word

Professor Martinez needs a state machine that will recognize the word 'mammal' when typed on a keyboard. Specifically, it must read any sequence of the letters m, a, and l (you may assume no other letters will be typed). It should move into a final state immediately after seeing 'mammal' (regardless of what letters came before those six in the sequence) and then remain in that final state as further characters come in. For efficiency, the state machine must be deterministic. Specifically, if you look at any state s and any action a, there is **exactly** one edge labelled a leaving state s.

Draw a deterministic state diagram that will meet his needs.